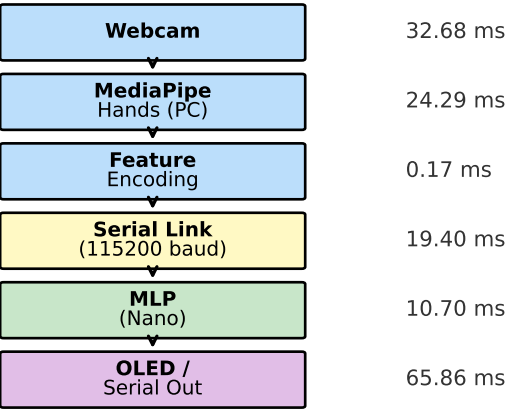


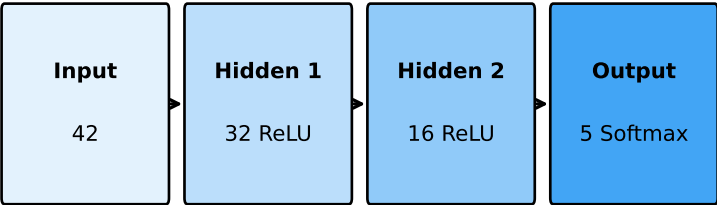
Design-Space Characterization of Layer-Streamed Neural Network Inference on the ATmega328P for Resource-Constrained ASL Landmark Classification

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End-to-End Pipeline (stage ms, measured)



MLP Architecture (1,989 parameters)

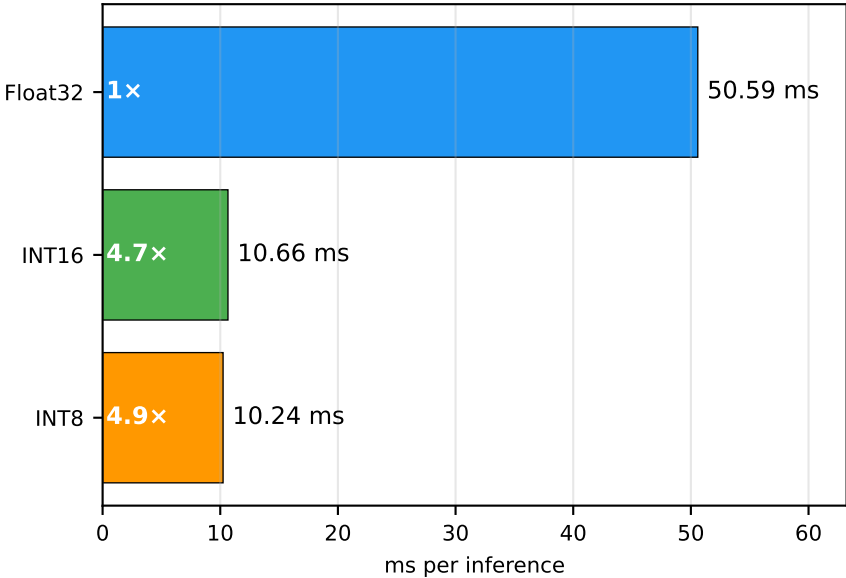


Streaming Ablation (conventional staging)

Topology	Static SRAM	Conventional verdict
small1	2,006 B	Compiles → crashes
small2	2,718 B	Compiler-rejected
current	3,430 B	Compiler-rejected
large1	4,854 B	Compiler-rejected
large2	6,278 B	Compiler-rejected

Layer-streaming: only executable strategy (5/5 topologies)

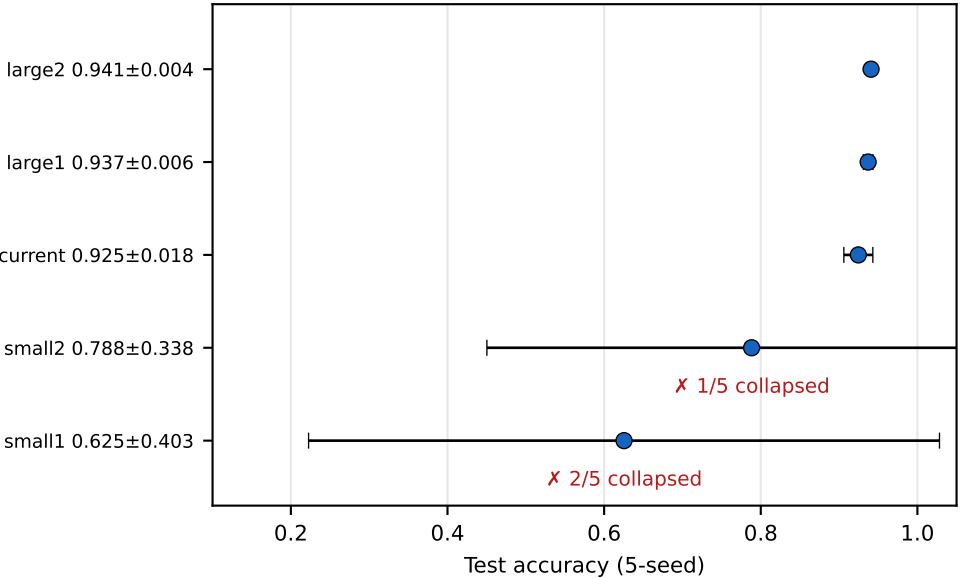
Inference Latency (infer_us, n=300)



Quantization Comparison (primary topology)

Metric	Float32	INT16	INT8
Accuracy	89.58%	89.58%	88.45%
Model Size	7,956 B	4,296 B	2,360 B
Firmware Size	27,326 B	23,778 B	21,884 B
Flash Usage	89%	77%	71%
Free RAM	826 B	950 B	930 B
Argmax Agreement	—	100%	98.0%
Avg Confidence	92.9%	93.9%	90.2%

Topology Replication (5 seeds each)



Key Findings: layer-streamed execution is the only weight-staging strategy that runs on this hardware (conventional staging fails at all 5 topologies) | INT16 matches Float32 accuracy (89.58%, 100% argmax) at 4.7x speedup with matching confidence

Five-seed replication: narrow topologies collapse (small1 2/5, small2 1/5 seeds); large1/large2 are the stable improvements | Scope: single-subject A-E hand-landmark classification